Eren C. Kızıldağ

Contact Information

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Research Interests

Applied probability, high-dimensional statistics, mathematics of data science, and theoretical computer science

Employment

University of Illinois Urbana-Champaign (July 2024 - Present)
Assistant Professor, Department of Statistics
Department of Electrical and Computer Engineering (Affiliate)

Columbia University (August 2022 - July 2024)
Distinguished Postdoctoral Fellow, Department of Statistics

Education

Massachusetts Institute of Technology

Ph.D. in Electrical Engineering and Computer Science (May 2022)
GPA: 5.0/5.0

- Advisor: Prof. David Gamarnik
- Thesis Title: Algorithms and Algorithmic Barriers in High-Dimensional Statistics and Random Combinatorial Structures

M.S. in Electrical Engineering and Computer Science (February 2017)
GPA: 5.0/5.0

- Thesis Title: Improved Magnetic Resonance Chemical Shift Imaging at 3 Tesla using a 32-channel Integrated RF-Shim Coil Array

Minor in Mathematics (February 2017)

Boğaziçi University (Istanbul, Turkey)

B.Sc. in Electrical and Electronics Engineering (June 2014)
GPA: 3.99/4.00

- 2nd rank in the university
- Specialized in Control Theory

Journals, Full Conference Papers & Preprints

Sharp Phase Transitions for the Overlap Gap Property
Eren C. Kızıldağ
SIAM Journal on Discrete Mathematics (To Appear), 2026+

Planted Random Number Partitioning Problem
Eren C. Kızıldağ
Algorithmic Learning Theory (ALT), 2026
(ALT 2026 Best Paper Award)

Large Average Subtensor Problem: Ground-State, Algorithms, and Algorithmic Barriers
Abhishek Hegade Kumbale Raveesha, Eren C. Kızıldağ
Algorithmic Learning Theory (ALT), 2026

Shattering in the Ising Pure p-Spin Model
David Gamarnik, Aukosh Jagannath, Eren C. Kızıldağ
Probability Theory and Related Fields, 2025

Information-Theoretic Guarantees for Recovering Low-Rank Tensors from

Symmetric Rank-One Measurements
Eren C. Kızıldağ
Algorithmic Learning Theory (ALT), 2025

Sharp Thresholds for the Overlap Gap Property: Ising p-Spin Glass and Random k-SAT
Eren C. Kızıldağ
Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM), 2025

Algorithmic Obstructions in the Random Number Partitioning Problem
David Gamarnik, Eren C. Kızıldağ
The Annals of Applied Probability, 2023

Geometric Barriers for Stable and Online Algorithms for Discrepancy Minimization
David Gamarnik, Eren C. Kızıldağ, Will Perkins, Changji Xu
Conference on Learning Theory (COLT), 2023

Stationary Points of a Shallow Neural Network with Quadratic Activations and the Global Optimality of the Gradient Descent Algorithm
David Gamarnik, Eren C. Kızıldağ, Ilias Zadik.
Mathematics of Operations Research, 2024
(Editor's Pick at MOR: 50 Papers for 50 Years)

Algorithms and Barriers in the Symmetric Binary Perceptron Model
David Gamarnik, Eren C. Kızıldağ, Will Perkins, Changji Xu
IEEE Symposium on Foundations of Computer Science (FOCS), 2022

Self-Regularity of Non-Negative Output Weights for Overparameterized Two-Layer Neural Networks
David Gamarnik, Eren C. Kızıldağ, Ilias Zadik.
IEEE Transactions on Signal Processing, 2022

Computing the Partition Function of the Sherrington-Kirkpatrick Model is Hard on Average
David Gamarnik, Eren C. Kızıldağ
The Annals of Applied Probability, 2021

Inference in High-Dimensional Linear Regression via Lattice Basis Reduction and Integer Relation Detection
David Gamarnik, Eren C. Kızıldağ, Ilias Zadik.
IEEE Transactions on Information Theory, 2021

Preprints and Permanent Manuscripts

Algorithmic Phase Transition for Large Independent Sets in Dense Hypergraphs
Abhishek Dhawan, Nhi U. Dinh, Eren C. Kızıldağ, Neelandri Maitra, Bayram A. Şahin
Under review, 2026+

Ground-State Energy of Ising Multipartite p-Spin Glasses for Large-p
Eren C. Kızıldağ, Bayram A. Şahin
Under review, 2026+

Optimal Hardness of Online Algorithms for Large Independent Sets
David Gamarnik, Eren C. Kızıldağ, Lutz Warnke
Under review, 2026+

Sharp Online Hardness for Large Balanced Independent Sets

Abhishek Dhawan, Eren C. Kızıldağ, Neeladri Maitra
Under review, 2026+

Symmetric Binary Perceptron with Random Labels: Capacity, Universality, and Overlap Gap Property
Eren C. Kızıldağ, Tanay Wakhare
Preprint, 2023

Neural Networks and Polynomial Regression. Demystifying the Overparametrization Phenomena
Matt Emschwiller, David Gamarnik, Eren C. Kızıldağ, Ilias Zadik
Permanent Manuscript, 2020

Conference Papers

A Random CSP with Connections to Discrepancy Theory and Randomized Trials
Eren C. Kızıldağ
IEEE International Symposium on Information Theory (ISIT), 2024

Symmetric Percepton with Random Labels
Eren C. Kızıldağ, Tanay Wakhare
International Conference on Sampling Theory and Applications (SampTA), 2023

The Random Number Partitioning Problem: Overlap Gap Property and Algorithmic Barriers
David Gamarnik, Eren C. Kızıldağ
IEEE International Symposium on Information Theory (ISIT), 2022

Self-Regularity of Output Weights for Overparameterized Two-Layer Neural Networks
David Gamarnik, Eren C. Kızıldağ, Ilias Zadik
IEEE International Symposium on Information Theory (ISIT), 2021

Computing the Partition Function of the Sherrington-Kirkpatrick Model is Hard on Average
David Gamarnik, Eren C. Kızıldağ
IEEE International Symposium on Information Theory (ISIT), 2020

High-Dimensional Linear Regression and Phase Retrieval via PSLQ Integer Relation Algorithm
David Gamarnik, Eren C. Kızıldağ
IEEE International Symposium on Information Theory (ISIT), 2019

Improved spiral chemical shift imaging at 3 Tesla using a 32-channel integrated RF-shim coil array
Eren C. Kızıldağ, Jason P. Stockmann, Borjan Gagoski, Bastien Guerin, P. Ellen Grant, Lawrence L. Wald, Elfar Adalsteinsson
International Society for Magnetic Resonance in Medicine (ISMRM), 2016
(Summa Cum Laude Award, Top 5%)

Other Experience

Simons Institute for the Theory of Computing, UC Berkeley
Computational Complexity of Statistical Inference Semester Program
Visiting Graduate Student (Fall 2021)

Simons Institute for the Theory of Computing, UC Berkeley
Probability, Geometry, and Computation in High Dimensions Semester Program
Visiting Graduate Student (Fall 2020)

Honors and Awards

Graduate School and Afterwards

- Best paper award, Algorithmic Learning Theory (ALT), 2026
- Mathematics of Operations Research, 50 Papers for 50 Years, Highlight Paper for 2025
- Columbia University, Distinguished Postdoctoral Fellowship at Statistics, 2022-24.
- University of Waterloo, Postdoctoral Fellowship in Probability & Mathematical Data Science, 2022 (Declined).
- MIT Graduate Student Council Travel Grant, 2019.
- *Summa cum laude* award (top 5%) in the 24th annual meeting of International Society for Magnetic Resonance in Medicine (ISMRM), 2016.

Undergraduate and Before

- Ranked 2nd in the graduating class of Bogazici University, 2014.
- Turkish Educational Foundation Supreme Success Scholarship, 2011-2014.
- Bogazici University Presidential Fellowship, 2010-2014.
- Turkish Ministry of Education Scholarship, 2010-2014.
- Semi-Finalist in Oyun'2010, 15th Turkish Intelligence Competition, 2010.
- Ranked 2nd in the graduating class of Ankara Science High School, 2010.[1]
- Ranked 11th among 1.8 million students in College Entrance Exam, 2010.
- Candidate in Turkish team for International Mathematical Olympiad (IMO), 2010.
- Silver Medal at International Silk Road Mathematical Competition, 2010.
- Bronze Medal at National Mathematical Olympiad, held by Scientific and Technological Research Council of Turkey (TUBITAK, Turkish equivalent of NSF), 2009.
- First rank at Mediterranean Mathematical Olympiad, 2009.
- First rank at Middle East Technical University Mathematical Competition, 2008.
- Bronze Medal at Mediterranean Mathematical Olympiad, 2007.
- Ranked 4th in the Centralized High School Entrance Exam, 2006.
- Gold Medal at National Junior Mathematical Olympiad, held by TUBITAK, 2006.

Selected Talks

- Algorithmic Learning Theory (ALT), February 2026
- International Conference on Randomization and Computation (RANDOM), August 2025
- Canadian Discrete & Algorithmic Mathematics Conference (CanaDAM), May 2025
- Bohrer Workshop, University of Illinois Urbana-Champaign, April 2025
- Probability Seminar, University of Illinois Urbana-Champaign, December 2024
- Northwestern University, Probability Seminar, November 2024
- Statistics Seminar, University of Illinois Urbana-Champaign, November 2024
- IEEE International Symposium on Information Theory, July 2024.
- Bocconi University, Workshop on Conceptual Challenges in AI, May 2024
- City University of New York (CUNY) Probability Seminar, April 2024
- ETH Zurich, Department of Mathematics, DACO Seminar, February 2024
- University of Illinois Urbana-Champaign, Statistics Seminar, February 2024
- Rensselaer Polytechnic Institute, Computer Science Colloquium, February 2024
- UMass Boston, Computer Science Seminar, February 2024
- University of Delaware, Mathematical Sciences Seminar, January 2024
- Bilkent University, Analysis Seminar, December 2023
- SUNY at Buffalo, Department of Mathematics Seminar, December 2023
- Rensselaer Polytechnic Institute Mathematical Sciences Colloquium, November 2023
- Stevens Institute of Technology, Mathematical Sciences Seminar, November 2023
- INFORMS Annual Meeting, October 2023.
- Columbia University, Statistics Student Seminar Series, October 2023.
- Columbia University, Applied Probability Seminar, October 2023.
- LU-UMN Joint Probability Seminar, September 2023.
- International Conference on Sampling Theory and Applications, July 2023.
- University of Chicago Theory Seminar, April 2023.
- Columbia University, Statistical Machine Learning Symposium, April 2023.

[1] A special high school that was modeled after Bronx High School of Science, with a curriculum tailored for gifted students.

- Penn/Temple Probability Seminar, March 2023.
- IEEE Symposium on Foundations of Computer Science (FOCS), November 2022.
- NYU Bruna Group Meeting, November 2022.
- Northeast Probability Seminar, November 2022.
- Columbia University CS Theory Seminar, November 2022.
- Columbia University Statistics Seminar, September 2022.
- Random Structures and Algorithms, August 2022.
- IEEE International Symposium on Information Theory, July 2022.
- MIT Statistics and Data Science Conference (SDSCon), April 2022.
- MIT SIAM Seminar, March 2022.
- UIC Combinatorics and Probability Seminar, March 2022.
- MIT LIDS and Statistics Tea Talk, February 2022.
- MIT LIDS Student Conference, January 2022.
- Stanford Information Theory Forum, January 2022.
- Stanford CS Theory Lunch, January 2022.
- Bilkent University, Electrical & Electronics Engineering Seminar, January 2022.
- Simons Institute at UC Berkeley, Graduate Seminar, November 2021.
- Cornell ORIE Young Researchers Workshop, October 2021.
- IEEE International Symposium on Information Theory, July 2021.
- MIT LIDS and Statistics Tea Talk, May 2021.
- MIT LIDS Student Conference, January 2021.
- MIT Machine Learning Tea Talks, July 2020.
- IEEE International Symposium on Information Theory, July 2020.
- Machine Learning at MIT Retreat, February 2020.
- IEEE International Symposium on Information Theory, July 2019.

Service

- Panelist for National Science Foundation (NSF)
- Seminar chair for UIUC Department of Statistics (since 2024)
- Session organizer at INFORMS Applied Probability Society Conference (2025)
- Member of the program committee for COLT (2023, 2024, 2025) and ALT (2024, 2025, 2026)
- Member of the organizing committee for Columbia University Statistical Machine Learning Symposium, 2023.
- Reading group organizer on Overlap Gap Property at Simons Institute Semester Program, *Computational Complexity of Statistical Inference*, Fall 2021.

Reviewer for

- Random Structures & Algorithms (RSA)
- Discrete Mathematics
- SIAM Journal on Discrete Mathematics (SIDMA)
- Physical Review X
- IEEE Information Theory Workshop (ITW)
- IEEE International Symposium on Information Theory (ISIT)
- Conference on Learning Theory (COLT)
- Algorithmic Learning Theory (ALT)
- Symposium on Theory of Computing (STOC)
- IEEE Symposium on Foundations of Computer Science (FOCS)
- International Colloquium on Automata, Languages, and Programming (ICALP)
- Mathematical Reviews for American Mathematical Society (AMS)

Teaching Experience

University of Illinois Urbana-Champaign

- STAT 553 - Probability and Measure I (Fall 2025)
- STAT 576 - Empirical Process Theory and Weak Convergence (Fall 2025)
- STAT 410 - Statistics and Probability II (Spring 2025)

Massachusetts Institute of Technology

Residential:

- Teaching Assistant for 6.436 Fundamentals of Probability (Fall 2019).

- Teaching Assistant for 6.344 Digital Image Processing (Spring 2019).
- Teaching Assistant for 6.S077 Introduction to Data Science (Spring 2018).
- Teaching Assistant for 6.437 Inference and Information (Spring 2017).
- Conducted weekly recitations and office hours, prepared and graded problem sets and exams.

Online:

- Teaching Assistant for 6.431x Probability—The Science of Uncertainty and Data, an edX-based MOOC with 2k enrollments. Answered and moderated forum posts, developed problem sets and exams (Fall 2018).
- Developed 6.431x Probability—The Science of Uncertainty and Data (Fall 2017) and 18.650x Fundamentals of Statistics (Summer 2018).